Atoms as Springs — Why Materials Stretch

Solid Mechanics · Lesson 1 of 5 · browser-based

1 · Bond | 2 · Hooke | 3 · Lattice | 4 · L-J — *concept sequence*

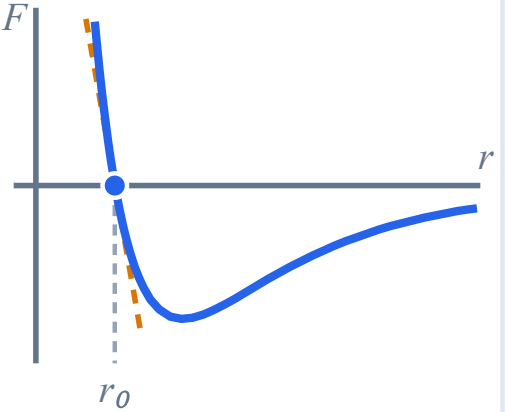
1 LEARN — guided math in plain language

$$F = -k\,x,\ x = r - r_0$$

$$U(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^{6}]$$

worked example (copper)

key takeaways



2 TRY THIS — prediction tasks

① *drag Separation — where does force cross zero?*

② *stretch past r_0 — does pull-back keep rising?*

3 EXPLORE — sliders drive the physics live

Controls

Separation 1.35

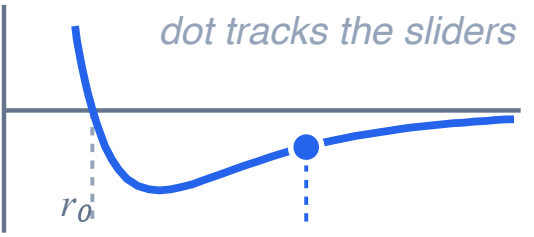
Strength ε 1.0

force F −1.1 ε/σ

stiff. k 57 ε/σ²

energy −0.30 ε





4 FEEDBACK — automatic validation

"pull-back weakening — the bond is near fracture"

iterate — revise intuition, test again

self-contained HTML — no install, runs on any device